

The Importance of Cells – How Cells Obtain Nutrients and Convert Them into Energy

Every living thing is made of cells. A single bacterium is one cell. A giant redwood tree contains billions of cells. A human being contains trillions. Despite their differences in size and shape, all cells share a common purpose: to obtain nutrients, convert them into usable energy, and use that energy to sustain life. Without this continuous process, no organism could grow, repair itself, respond to its environment, or reproduce. Cells are not passive containers. They are active chemical factories that constantly take in raw materials, break them down, build new molecules, and release energy. Understanding how cells obtain nutrients and convert them into energy is fundamental to understanding life itself.

This reading material describes how cells obtain nutrients from food and convert them into energy through cellular processes to sustain life in plants and animals.

Part 1: Cells as the Basic Unit of Life

What Is a Cell?

A **cell** is the smallest structural and functional unit of life. Every cell is surrounded by a **cell membrane** that controls what enters and exits. Inside, the cell contains **cytoplasm**, a jelly-like fluid where chemical reactions occur, and **organelles**, specialized structures that perform specific jobs.

Why Cells Are Important

Reason	Explanation
Basic unit of structure	All living things are made of one or more cells
Basic unit of function	All life processes occur within cells
Site of energy production	Cells convert nutrients into usable energy
Site of growth and repair	Cells divide to produce new cells
Site of heredity	Cells contain DNA that passes traits to offspring
Response to environment	Cells detect and respond to changes

Key Organelles Involved in Energy Processes

Organelle	Function	Found In
Cell membrane	Controls movement of substances in and out	All cells
Cytoplasm	Site of many chemical reactions	All cells
Mitochondrion	Site of cellular respiration; produces ATP	Plant and animal cells
Chloroplast	Site of photosynthesis; captures light energy	Plant cells only

Organelle	Function	Found In
Ribosome	Builds proteins	All cells
Vacuole	Stores water, nutrients, and wastes	Plant cells (large); animal cells (small)

Part 2: How Cells Obtain Nutrients

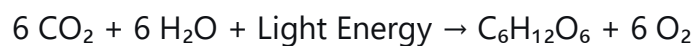
Plants: Making Food from Sunlight

Plants are **autotrophs**, meaning they produce their own food. They do not consume other organisms. Instead, they use sunlight, water, and carbon dioxide to make glucose through **photosynthesis**.

Photosynthesis Overview

Location: Chloroplasts in leaf cells

Equation:



Inputs of Photosynthesis

Input	Source	How It Enters the Cell
Carbon dioxide (CO ₂)	Air	Diffuses through stomata into leaf cells
Water (H ₂ O)	Soil	Absorbed by roots; transported through xylem to leaves

Input	Source	How It Enters the Cell
Light energy	Sun	Absorbed by chlorophyll in chloroplasts

Outputs of Photosynthesis

Output	Where It Goes
Glucose (C ₆ H ₁₂ O ₆)	Used for energy, growth, or stored as starch
Oxygen (O ₂)	Released into the air through stomata

How cells obtain nutrients in plants:

1. Root cells absorb water and minerals from the soil.
2. Water travels through xylem vessels to leaf cells.
3. Carbon dioxide enters leaf cells through stomata.
4. Chloroplasts capture light energy and use it to combine CO₂ and H₂O into glucose.
5. Glucose is used by the cell for energy or converted into other molecules for growth.
6. Oxygen is released as a waste product.

Animals: Obtaining Nutrients from Food

Animals are **heterotrophs**, meaning they cannot make their own food. They must consume other organisms to obtain nutrients.

The Journey of Nutrients from Food to Cells

Step	What Happens	Where It Occurs
1. Ingestion	Food is taken into the body	Mouth
2. Digestion	Food is broken down into small molecules	Stomach and intestines
3. Absorption	Small molecules pass into the bloodstream	Small intestine
4. Transport	Blood carries nutrients to cells	Circulatory system
5. Uptake	Nutrients enter cells through the cell membrane	All body cells

How Nutrients Enter Cells

Nutrient	How It Enters the Cell
Glucose	Facilitated diffusion or active transport through protein channels
Amino acids	Active transport through carrier proteins
Oxygen	Simple diffusion through the cell membrane
Water	Osmosis through aquaporins

How cells obtain nutrients in animals:

1. Food is eaten and broken down mechanically and chemically.
2. Digestive enzymes break large molecules into glucose, amino acids, and fatty acids.
3. These small molecules are absorbed into the blood through the small intestine.
4. The circulatory system delivers nutrients to every cell in the body.
5. Nutrients cross the cell membrane and enter the cytoplasm.
6. Inside the cell, nutrients are processed to release energy or build new molecules.

Comparison: How Plants and Animals Obtain Nutrients

Feature	Plants	Animals
Type	Autotroph	Heterotroph
Source of nutrients	Sunlight, water, carbon dioxide	Food (other organisms)
Site of food production	Chloroplasts	Not applicable
How nutrients reach cells	Xylem and phloem transport	Circulatory system transports
Storage form of energy	Starch	Glycogen
Key organelle	Chloroplast	Mitochondrion (for energy release)

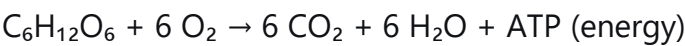
Part 3: How Cells Convert Nutrients into Energy

Cellular Respiration: The Universal Energy Process

Once nutrients enter a cell, they must be broken down to release energy. This process is called **cellular respiration**. It occurs in both plant and animal cells, and it is the primary way cells obtain usable energy.

Location: Cytoplasm (glycolysis) and mitochondria (Krebs cycle and electron transport chain)

Equation:



What Is ATP?

ATP (adenosine triphosphate) is the energy currency of the cell. It stores energy in the bonds between its phosphate groups. When a cell needs energy, it breaks the last phosphate bond, releasing energy and forming ADP (adenosine diphosphate). ATP is constantly being used and regenerated.

Molecule	Full Name	Role
ATP	Adenosine triphosphate	Stores and transfers energy
ADP	Adenosine diphosphate	Formed when ATP releases energy
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	Primary fuel for cellular respiration

The Four Stages of Cellular Respiration

Stage	Location	Input	Output	ATP Produced
1. Glycolysis	Cytoplasm	Glucose	2 pyruvate, 2 NADH	2 ATP (net)
2. Formation of Acetyl CoA	Mitochondrial matrix	Pyruvate	Acetyl CoA, CO ₂ , NADH	0
3. Citric Acid Cycle (Krebs Cycle)	Mitochondrial matrix	Acetyl CoA	CO ₂ , NADH, FADH ₂	2 ATP
4. Electron Transport Chain and Chemiosmosis	Inner mitochondrial membrane	NADH, FADH ₂ , O ₂	H ₂ O, ATP	~34 ATP

Total ATP per glucose molecule: Approximately **36–38 ATP**

Aerobic vs. Anaerobic Respiration

Feature	Aerobic Respiration	Anaerobic Respiration (Fermentation)
Oxygen required?	Yes	No
Location	Cytoplasm + mitochondria	Cytoplasm only

ATP produced per glucose	36–38 ATP	2 ATP
Byproducts	CO ₂ and H ₂ O	Lactic acid (animals) or ethanol + CO ₂ (yeast)
When used	Most of the time; during normal activities	During intense exercise or when oxygen is limited

How Plants and Animals Use Cellular Respiration

Organism	How It Obtains Glucose	How It Uses Cellular Respiration
Plants	Produces glucose through photosynthesis	Breaks down glucose to power growth, transport, and reproduction
Animals	Obtains glucose from digested food	Breaks down glucose to power movement, thinking, and body functions

Key insight: Plants also perform cellular respiration. They use the glucose they produce during photosynthesis to fuel their own life processes.

Part 4: Comparing Photosynthesis and Cellular Respiration

Photosynthesis and cellular respiration are opposite processes that work together in a cycle.

Feature	Photosynthesis	Cellular Respiration
Purpose	To store energy in glucose	To release energy from glucose
Location	Chloroplasts	Cytoplasm and mitochondria
Inputs	CO ₂ , H ₂ O, light energy	C ₆ H ₁₂ O ₆ , O ₂
Outputs	C ₆ H ₁₂ O ₆ , O ₂	CO ₂ , H ₂ O, ATP
Organisms	Plants, algae, some bacteria	All living organisms
Energy conversion	Light energy → chemical energy	Chemical energy → usable energy (ATP)

The cycle:

Photosynthesis produces glucose and oxygen → Cellular respiration uses glucose and oxygen to produce ATP, carbon dioxide, and water → Carbon dioxide and water are used again for photosynthesis

This cycle connects plants and animals. Plants produce oxygen and glucose. Animals consume oxygen and glucose. Both release carbon dioxide and water, which plants use to continue photosynthesis.

Part 5: Why This Matters – Sustaining Life

How Cellular Energy Sustains Life in Plants

Life Process	How ATP Is Used
Growth	Building new cells and tissues
Transport	Moving water and nutrients through xylem and phloem
Reproduction	Producing flowers, seeds, and fruits
Defense	Producing chemicals to deter herbivores
Response	Growing toward light (phototropism)

How Cellular Energy Sustains Life in Animals

Life Process	How ATP Is Used
Movement	Muscle contraction for walking, running, flying
Thinking	Nerve cells transmitting electrical signals
Digestion	Producing enzymes and absorbing nutrients

Life Process	How ATP Is Used
Circulation	Heart pumping blood throughout the body
Repair	Replacing damaged cells and healing wounds
Reproduction	Producing gametes and supporting offspring

What Happens Without Energy?

Situation	Consequence
No glucose available	Cell cannot produce ATP; cell function stops
No oxygen available	Aerobic respiration stops; cell switches to anaerobic respiration (less ATP)
No ATP production	Cell cannot maintain itself; cell dies
Cell death	Tissue fails; organ fails; organism dies

Real-World Connections

Scenario	Cellular Explanation
A plant wilts in drought	Lack of water stops photosynthesis; less glucose means less ATP for growth and transport

Scenario	Cellular Explanation
A runner feels muscle fatigue	Oxygen supply cannot meet demand; cells switch to anaerobic respiration, producing lactic acid
A person feels tired after skipping breakfast	Low blood glucose means less fuel for cellular respiration; less ATP for brain and muscles
A plant grows toward a window	Cells on the shaded side elongate more; ATP powers the growth response
A fever develops during infection	Immune cells increase cellular respiration to fight pathogens; body temperature rises

Key Terms

Term	Definition
Cell	The basic structural and functional unit of life
Autotroph	An organism that produces its own food (e.g., plants)
Heterotroph	An organism that obtains food by consuming other organisms (e.g., animals)
Photosynthesis	The process by which plants convert light energy into chemical energy stored in glucose
Cellular respiration	The process by which cells break down glucose to release energy in the form of ATP
ATP	Adenosine triphosphate; the energy currency of the cell
ADP	Adenosine diphosphate; formed when ATP releases energy
Glycolysis	The first stage of cellular respiration; occurs in the cytoplasm; splits glucose into pyruvate
Krebs cycle	The third stage of cellular respiration; occurs in the mitochondrial matrix; produces ATP, NADH, FADH ₂

Term	Definition
Electron transport chain	The final stage of cellular respiration; produces most of the ATP
Aerobic respiration	Respiration that requires oxygen; produces 36–38 ATP per glucose
Anaerobic respiration	Respiration without oxygen; produces 2 ATP per glucose
Chloroplast	Organelle where photosynthesis occurs
Mitochondrion	Organelle where cellular respiration occurs
Glucose	$C_6H_{12}O_6$; the primary fuel for cellular respiration

Review Questions

1. Explain why cells are considered the basic unit of life.
2. Describe how plant cells obtain the raw materials needed for photosynthesis. Include the sources of carbon dioxide, water, and light energy.
3. Explain how animal cells obtain glucose from food. Include the roles of the digestive system and circulatory system.
4. Write the balanced chemical equation for cellular respiration and identify the inputs and outputs.
5. What is ATP and why is it important for cells?
6. List the four stages of aerobic cellular respiration and state the location of each.
7. Differentiate between aerobic and anaerobic respiration in terms of oxygen requirement, ATP production, and byproducts.
8. Explain why plants also perform cellular respiration even though they produce their own glucose through photosynthesis.
9. Study the scenario below and answer the questions that follow.

A student observes that a potted plant placed in a dark closet for one week turns yellow and stops growing. When moved back to a sunny windowsill, it gradually turns green and begins to grow again.

- (a) What process stopped when the plant was in the dark?
- (b) Why did the plant stop growing?
- (c) What happened when the plant was returned to the light?
- (d) Explain how this demonstrates the importance of cells converting nutrients into energy.

10. Application question: A marathon runner eats a large pasta meal the night before a race and drinks water throughout the event. During the final kilometer, the runner's muscles begin to fatigue and burn.

- (a) Explain how the pasta meal provided energy for the runner's cells.
- (b) Describe the process by which cells converted the pasta into usable energy before the race.
- (c) Explain why the runner's muscles began to fatigue during the final kilometer.
- (d) What would happen to the runner's cells if oxygen were completely unavailable?